Guided Assignment 3

From vector computations to neural memory

PN3 · Nengo tutorial · Lecture 6

Purpose. Build the written reasoning needed before the Test 3 practice paper. PN3 explicitly prepares for Test 3; this assignment follows its core tasks. It is untimed learning practice, not an exam prediction. Lecture extensions appear only as optional work.

How to work. Predict, calculate, then explain. Use paper, notes, and a basic calculator through Stage 8; no programming or Nengo installation is required. Use the space provided for a first sketch and separate paper for longer work. Attempt each part before opening the companion's hints; check solutions after a stage, then close them and redo missed reasoning.

Pacing. Three or four flexible sittings: Stages 1–2; Stages 3–4; Stages 5–6; Stages 7–9. Combine the first two if comfortable. Allow about 45–75 minutes per sitting, longer for unfamiliar prerequisites. Keep Stage 9 closed-notes and save the practice paper until afterward.

Conventions. Time is in seconds unless labelled otherwise; rates are in Hz. Vectors are columns. Radius R describes a trained domain, not automatic clipping. A d-dimensional radius usually describes a ball, not a coordinatewise box. State $x(0) = 0$ for all ideal integrals. Probe filtering changes what is recorded; a connection filter changes the signal reaching the network.

1. Recover the representation chain

Skill: Separate values, neurons, currents, and decoders.

1a. Prerequisite. In $J_i = \alpha_i \mathbf{e}_i^T(\mathbf{x}/R) + \beta_i$, name the roles of $\mathbf{x}, R, \mathbf{e}_i, \alpha_i, \beta_i$. For 100 neurons representing a 2-D vector, how many represented coordinates and how many currents are there?

1b. Calculate. With $R = 2$, $\mathbf{e} = (3/5, 4/5)^T$, $\mathbf{x} = (1, 1)^T$, $\alpha = 2$, $\beta = 0.3$, find the normalized projection and current. Does a positive current itself specify a represented coordinate?

1c. Explain. With n neurons and N samples, activities form $A \in \mathbb{R}^{n \times N}$. State the identity decoder's shape for a 2-D output. Explain what Nengo automates when it builds a connection, and why a decoded estimate need not equal the target exactly.

Before moving on: You distinguish 100 neuron activities from a 2-coordinate represented value.

Working space / questions to revisit

2. Assemble vector computations

Skill: Translate a vector equation into transforms, ranges, and neural weights.

PN3 asks for $\mathbf{w} = \mathbf{x} - 3\mathbf{y} + 2\mathbf{z} - 2\mathbf{q}$, using five 2-D populations. First predict signs and rough sizes, then calculate.

2a. Guided. For $\mathbf{x} = (0.5, 1)^T$, $\mathbf{y} = (0.1, 0.3)^T$, $\mathbf{z} = (0.2, 0.1)^T$, $\mathbf{q} = (0.4, -0.2)^T$, compute all four contributions and their sum. Draw the five populations and label every incoming transform to W as a 2×2 matrix.

2b. Transfer. Replace y_1 and q_1 by $s(t) = \sin(4\pi t)$; keep other components as above. Derive both coordinates of $\mathbf{w}(t)$, its period, and the smallest symmetric radius containing its entire ideal trajectory. Is radius 1 sufficient for $\mathbf{x}$? Why is checking each coordinate separately insufficient?

2c. From transforms to weights. Let the X population have n neurons, identity decoder $D_X \in \mathbb{R}^{2\times n}$, and filtered activities $\mathbf{a}_X$. Define E_W with row $\alpha_i \mathbf{e}_i^T / R_W$. Write the $X \to W$ weight matrix for transform T_X, its shape for m receiving neurons, and one receiving neuron's current including all four branches. Where is its neuronal bias added?

2d. Explain. Compare a hand-built implementation and a Nengo implementation of this diagram. Give three things that must match for their decoded traces to agree closely. Would exact agreement at every instant be expected from independently sampled spiking populations?

Before moving on: You can calculate the target before diagnosing the neural approximation.

Working space / questions to revisit

3. Radius and a fair static accuracy comparison

Skill: Predict radius scaling while stating what is held fixed.

Use PN3's 100-neuron, 1-D ensemble, normalized intercepts in $[-1, 1]$, maximum rates in $[100, 200]$ Hz, $\tau_{RC} = 20$ ms and $\tau_{ref} = 2$ ms. A normalized intercept is the threshold projection z_i in x/R, not a physical value fixed across radii.

3a. Guided. A positive-encoder neuron has normalized intercept $z = 0.25$. Give its physical intercept at radii $0.5, 1, 2, 4$. At corresponding normalized positions $r = x/R$, should its current and firing rate change when only the radius changes and normalized tuning parameters stay fixed?

3b. Calculate symbolically. Fit identity decoders using the same normalized samples r_k, activity matrix A, and regularizer $\lambda > 0$ at every radius. Targets are $X_R = RX_1$ and

$$D_R = X_R A^T (AA^T + \lambda I)^{-1}.$$

Derive D_R in terms of D_1 and the resulting relation between physical-unit RMSEs. If RMSE at radius 1 is 0.03, complete a prediction table for radii $0.5, 1, 2, 4$. What happens to RMSE divided by radius?

3c. Explain the comparison. Why does testing all radii on the same physical interval $[-0.5, 0.5]$ no longer establish that scaling law? List the controls for a fair normalized-domain comparison, including neuron realization, sample positions, solver/noise convention, and target function.

3d. Transfer. To represent $p = ab$ for all $(a, b) \in [-1, 1]^2$, why is a joint population useful? Find a sufficient radius for that joint domain. Explain how enlarging a radius can cover more values while increasing absolute identity decoding error under the controlled scaling experiment.

Before moving on: You state a domain and a sampling convention whenever claiming a radius–RMSE relation.

Working space / questions to revisit

4. Change LIF time constants without changing the question

Skill: Distinguish fixed-current effects from recalibrated tuning-curve sweeps.

The steady LIF rate is zero for $J \leq 1$; for $J > 1$,

$$G(J) = [\tau_{ref} - \tau_{RC} \ln(1 - 1/J)]^{-1} .$$

A positive-encoder neuron has $J = \alpha r + \beta$, normalized intercept $z = 0$, and maximum rate $a_{max} = 100$ Hz at $r = 1$.

4a. Predict with fixed gain/bias. At a fixed suprathreshold current J, predict the effect of increasing τ_{ref} and of increasing τ_{RC}. State the asymptotic rate ceiling as $J \to \infty$.

4b. Guided calibration. Solve $G(J_{max}) = a_{max}$ for J_{max}. Use $J(z) = 1$ and $J(1) = J_{max}$ to derive α and β for a general $z < 1$. Use seconds. For $z = 0$, $\tau_{RC} = 0.020$, $\tau_{ref} = 0.002$, calculate approximate J_{max}, α, β. You may use $e^{-0.4} \approx 0.6703$.

4c. Recalibrate. Increase τ_{ref} to 0.005 s, keeping z and a_{max} fixed. Recalculate using $e^{-0.25} \approx 0.7788$. Sketch the original and new tuning curves, marking the common threshold and endpoint. Must the interior curves be identical? Why is simply lowering the endpoint rate the wrong prediction for this experiment?

4d. Design a paper sweep. Describe controls and measurements for PN3's $\tau_{ref} = 1$–5 ms and $\tau_{RC} = 10$–100 ms sweeps. Explain why gains/biases and decoders must be recomputed. At $\tau_{ref} = 5$ ms, can a neuron have a finite-current maximum rate of exactly 200 Hz? Can the algebra alone guarantee a monotonic population RMSE trend across these sweeps?

Before moving on: You know whether a comparison fixes currents or fixes intercepts and maximum rates.

Working space / questions to revisit

5. Predict feedforward steps and recorded traces

Skill: Separate the desired map from connection and probe filtering.

PN3 connects a 100-neuron scalar ensemble X to a 50-neuron scalar ensemble Y, with a 10 ms connection synapse and 10 ms probe synapses. Let the source represent the ideal pulse $x = 1$ for $0.1 < t < 0.4$ s and zero otherwise. For this calculation, ignore source representation errors and use no extra filter before X.

5a. Guided. Draw the input, X, Y, connection filter, and probes. For identity, give the steady values before, during, and after the pulse. Sketch raw input and both recorded outputs. Why can the Y probe look slower even when both probes use 10 ms?

5b. Calculate. A unit step reaches one first-order filter at time t_0. Its output is $1-e^{-\Delta/\tau}$, $\Delta = t-t_0 \geq 0$. Derive or verify the response through two identical cascaded filters,

$$1 - (1 + \Delta/\tau)e^{-\Delta/\tau}.$$

Compare the two values at $\Delta = \tau$, using $e^{-1} \approx 0.368$. Write the identity pulse response as the difference of two shifted step responses.

5c. Affine transfer. Now Y should compute $y = 1 - 2x$. Find the three ideal plateau values. Specify either a function decoder for the entire affine map or a -2 transform plus an explicit constant input. Why can a scalar linear transform alone not supply the $+1$? Is the receiving neurons' bias automatically that represented constant?

5d. Initialization and diagnosis. If the affine target is present from simulation time zero but filters start at zero, should the recorded output start at 1? Explain why a correct plateau with rounded transitions suggests a different issue from an incorrect plateau after settling. Would increasing neuron count necessarily eliminate connection/probe lag?

Before moving on: You can sketch a mathematical target and the filtered observation as distinct traces.

Working space / questions to revisit

6. Turn a synapse into a desired dynamical system

Skill: Derive the recurrence transform and check dimensions and units.

For matched first-order input and recurrent synapses with time constant τ, the ideal represented-state model is

$$\tau\dot{\mathbf{x}} = -\mathbf{x} + \mathbf{f}'(\mathbf{x}) + \mathbf{g}'(\mathbf{u}).$$

Primes here label implemented connection functions, not derivatives. The target is $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$.

6a. Guided derivation. Substitute $\mathbf{f}' = A'\mathbf{x}$ and $\mathbf{g}' = B'\mathbf{u}$. Match coefficients to derive A' and B'. If $\mathbf{x} \in \mathbb{R}^d$ and $\mathbf{u} \in \mathbb{R}^p$, give all four matrix shapes. Why must the identity matrix occur only in the state transform?

6b. Integrator. For d-dimensional integration of a d-dimensional input, state A, B, A', B'. At recurrent $\tau = 50$ ms, give the scalar connection transforms. Predict what happens if the input transform is set to 1 instead of τ in the matched-filter model.

6c. Transfer to leak. For $\dot{x} = -2x + u$, with $\tau = 0.05$ s, find the transforms. With $u = 0$ and $x(0) = 0.6$, write the ideal decay and its time constant. Explain why recurrent transform 0.9 does not mean multiplying the state by 0.9 once per second.

6d. Feedback distortion. An intended integrator instead decodes recurrent feedback $f'(x) = x + \epsilon(x)$. Derive $\dot{x}$ with the correct input transform. For $u = 0$ and constant $\epsilon = 0.001$, calculate the drift per second at $\tau = 0.05$ s. If $\epsilon(x)$ crosses zero with negative slope, what local behavior could replace perfect memory?

Before moving on: You can derive transforms from the desired differential equation rather than recalling a feedback gain.

Working space / questions to revisit

7. Integrate the input, then inspect the neural result

Skill: Use definite integrals, continuity, and radius checks to predict traces.

Assume $x(0) = 0$ and $x(t) = \int_0^t u(t')\,dt'$. Each input below is a separate experiment. Predict shapes before integrating; label event times and final values.

7a. Guided pulse. For $u = 0.9$ on $[0.04, 1.0)$ and zero otherwise, complete the three branches for $t < 0.04$, $0.04 \le t < 1$, and $t \ge 1$. Check continuity and compute $x(1.5)$. Why does state not return to zero when input stops?

7b. Transfer pulse. Shorten the same input to $[0.04, 0.16)$. Write all branches and the held value. Compare the area with part (a). Does a smaller held value guarantee smaller relative error in a neural implementation?

7c. Ramp. The input rises linearly from 0 to 0.9 between $t = 0$ and 0.45 s and is zero afterward. First write $u(t)$ with units consistent with seconds, then find the definite integral on both intervals. Check your final value using triangle area. Distinguish a ramp input from a ramp state.

7d. Sine. For $u = 5\sin(5t)$, derive the definite integral, its period and its range. Does radius 1 cover the ideal state? Explain why $-\cos(5t)$ alone fails the initial condition. Choose a sufficient symmetric state radius and mention the accuracy cost identified in Stage 3.

Before moving on: Your integral is continuous, satisfies $x(0) = 0$, and differentiates back to the input away from switches.

Working space / questions to revisit

8. Diagnose dynamics and control the feedback

Skill: Explain residual errors and design a variable-feedback integrator.

PN3 uses a 200-neuron integrator, recurrent synapse 50 ms, input synapse 5 ms, and probe synapse 10 ms. Use the recurrent time constant when choosing transforms, as PN3 directs.

8a. Diagnose. Compare spiking LIF and LIFRate versions. Predict which visible fluctuations should usually decrease in rate mode. Name three deviations from the ideal integral that can remain. Explain why rate mode does not turn an approximate recurrent decoder into perfect memory.

8b. Unequal filters. Which assumption of Stage 6 is violated? Why is substituting the input's 5 ms for τ in B' not justified by that derivation? Explain why a 10 ms probe can smooth a trace without repairing state drift. Propose a matched-filter comparison and a repeated-seed comparison that would help distinguish mechanisms.

8c. Controlled integrator: define the model. Here feedback gain is an external input $c(t)$, in s^{-1}, and the target is explicitly $\dot{x} = u + c(t)x$. With matched $\tau = 0.05$ s, derive the input and feedback functions. For $u = 0$, compare $c = 0, -2, +2$ starting from $x(0) = 0.4$. Which setting holds memory ideally? For constant $u = 0.3$ and $c = -2$, find the equilibrium and whether nearby states approach it.

8d. Two populations. Design a state population X and a joint population C representing $(x, \tilde{c})$, where $\tilde{c} = c/c_0$ and $c_0 = 2\,\mathrm{s}^{-1}$. Route x and $\tilde{c}$ into separate coordinates of C; decode the appropriate feedback from C into X. Label functions, dimensions, domain coverage, and where the input enters. Why is this an approximation to the desired feedback when the extra routing connection has its own dynamics? State a paper demonstration using a pulse followed by changes of c.

Optional lecture extension. For $\dot{x} = u - x^3$, write the transformed feedback and input functions for matched synapses. With $u = 0$, use the sign of $\dot{x}$ to predict whether small positive and negative states move toward zero. No programming required.

Before moving on: You can distinguish variable feedback cx from multiplying only the external input by a gate.

Working space / questions to revisit

9. Independent readiness check

Skill: Retrieve the core reasoning with the scaffolding removed.

Close notes and solutions. Use a fresh sheet and show equations, labelled sketches, and brief explanations. This is untimed; keep exact expressions where convenient. Use a calculator only after establishing the equations.

9a. Draw five 2-D populations computing $w = x - 3y + 2z - 2q$, labelling transforms. For $x = (0.2, 0.4)$, $y = (0, 0.1)$, $z = (0.3, 0)$, $q = (0.1, -0.2)$, find w and a sufficient radius for that output. Give the shape of weights from 80 source neurons to 60 receiving neurons.

9b. Under a normalized-domain radius sweep, identity RMSE at $R = 2$ is 0.08. Predict it at $R = 0.5$ and state assumptions. Explain what changes in a LIF time-constant sweep that preserves intercepts and maximum rates; give the calibration equations and the feasibility condition on the maximum rate.

9c. A scalar source represents a pulse of height 1 for $0.2 < t < 0.5$. The target is $1 - 2x$. Give ideal plateaus, a valid connection design, and qualitative traces when connection and probe synapses are each 10 ms. Explain the receiving neuronal bias's role.

9d. Derive A', B' for $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ under matched synapses. Specialize to $\dot{x} = u$ at $\tau = 0.04$ s. For input $u = 0.6$ on $[0.1, 0.4)$ and zero otherwise, write the definite integral in all intervals. Separately integrate $u = 2\sin(4t)$ from zero and find its range.

9e. A rate integrator drifts after a pulse; the spiking version also jitters. Explain possible sources of each, the effect of probe smoothing, and the limitation of unequal input/recurrent synapses. Explain why more neurons do not guarantee lower total RMSE in every run.

9f. For $\dot{x} = u + cx$, give feedback/input functions at $\tau = 0.04$ s, a two-population diagram for external c, and the predicted zero-input behavior for $c = 0$ and $c = -1\,\mathrm{s}^{-1}$. Explain why a fixed linear transform cannot compute cx for independently varying c, x.

Readiness rule. Mark every part *independent*, *needed a hint*, or *not yet clear*. Revisit the relevant stage and redo a nearby problem before retrying any part needing help. Attempt the practice paper when all six can be answered independently with consistent units, dimensions, and explanations. This is a learning criterion, not a guarantee about the real test.

Tutoring in this chat. Send one part, your reasoning, and confidence. Ask for a small hint, a reasoning check, or a transfer question before requesting the worked answer.

Working space / questions to revisit